

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: ARABICMMLU | **Context:** Non-Arab Tourists and Expats in Eight Arabic-Speaking Countries (ArabicMMLU Deployment)

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Authentic ground-up sourcing from school exams across eight countries [Q1, Q26] is a substantial strength over translation-based prior work [Q5, Q6], and Palestine being a top-3 source country is materially better than the scaffold anticipated [WEB-3, WEB-4]. However, content distribution is severely unequal: Jordan contributes 6,000+ questions while some countries contribute as few as 100 [Q89]; Morocco lacks annotators and Maghrebi school-curriculum coverage [DATASET-D5]; Kuwait is effectively absent [Limitations.5]; and contested historical content is framed through Jordanian curriculum perspective [DATASET-D1, DATASET-D2, DATASET-D3]. OCR corruption in some Egyptian university items [DATASET-D23, DATASET-D24] introduces construct-irrelevant variance. Islamic Studies content is presented prescriptively from an internal Muslim perspective [DATASET-D14, DATASET-D35], misaligned with the non-Muslim tourist target population. Per the elicitation, pan-Arab framing is acceptable [A2], but the system should flag country-specific content — yet the benchmark's ground-truth treats Jordanian framing as the silent default for civic and historical questions [DATASET-D4].

ID	Evidence
Q1	"we present ArabicMMLU, the first multi-task language understanding benchmark for the Arabic language, sourced from school exams across diverse educational levels in different countries spanning North Africa, the Levant, and the Gulf regions." (p.1)
Q26	"The questions are sourced from eight different countries in North Africa (Morocco and Egypt), the Levant (Jordan, Palestine, and Lebanon), and the Gulf (UAE, Kuwait, and KSA)." (p.3)
Q5	"Although large language models (LLMs) such as GPT-3.5 (Ouyang et al., 2022), BLOOMZ (Muennighoff et al., 2022), and Jais (Sengupta et al., 2023) have been pretrained with substantial coverage of Modern Standard Arabic (MSA), their reasoning and knowledge assessments are primarily conducted using datasets translated from English to Arabic (Sengupta et al., 2023; Huang et al., 2023), which means there is limited capacity to evaluate content specific to Arabic." (p.1)
Q6	"This reliance on translation systems not only demonstrates an Anglocentric approach (Ramesh et al., 2023; Talat et al., 2022) but also potentially introduces errors and biases." (p.1)
Q89	"ArabicMMLU does not represent all Arabic countries equally. For example, we have collected over 6K multiple-choice questions from Jordan, while other countries are represented with only 100 questions or, in some cases, not at all." (p.9)
WEB-3	Palestine confirmed top-3 source country — yet no Palestinian annotators in pool (https://github.com/mbzuai-nlp/ArabicMMLU)
WEB-4	HuggingFace dataset card confirms Palestine major contributor (https://huggingface.co/datasets/MBZUAI/ArabicMMLU)
D5	ين طول اعافد اب فكل كم الم ايمو كالح اطل سل او لدعل ريزو نم ... فانئئست ال ا لج ائدتبي (Professional) items are from Morocco (Moroccan procedural law)
D1	دحو مامت ا: ني طس ل ف ي ا ه ا ع ا ط ا و ا ي ن و ي ه ص ل ا ك ر ح ل ا نم ل و ا ل ا ه ل ا د ب ع ك ل م ا ف ق ا و م نم تس ل ا ي ت ا ل ا نم د ح ا و هض فر / ني طس ل ف ل ا ا م ا ل ا ق و د ن ص م ع د ا ل ا ب ر ع ل ا ق و ع د / ق ا ر ب ل ا ط ئ ا ح ا ل ا ا ر م د و ه ي ل ا ا ط ع ا هض فر / م 1950 ا ع ا ن ي ت ف ض ل ا ا ع ا ط ا ق ا ض فر ر و ف ل ب د ع و و و ك ي ب س ك ي ا س ق د ه ا ع — Jordan HS history question about King Abdullah I's positions on Zionism and Palestine — framed from Jordanian curriculum perspective
D2	ن ب ل ص ي ف ر ي م ا ل ا ٢٠ / ن ي س ح ل ا ن ب ل ل ا د ب ع ر ي م ا ل ا ١٠: س ي ر ا ب ي ف ا ف ل ح ل ا ه د ق ع ي ذ ل ا ح ل ص ل ا ر م ت و م ف ي ب ر ع ل ا ____ ل ث م ن ي س ح ل ا ٣ / ن ي س ح ل ا ٤ / ي ل ع ن ب ن ي س ح ل ا ف ي ر ش ل ل ا ٣ / ن ي س ح ل ا — Jordan civics HS: who represented Arabs at Paris Peace Conference — Jordanian Hashemite framing
D3	ب ا د ت ن ا ل ا ك ف د و ي ق ن م د ا ل ب ل ا ر ر ح ت: ١٩٢٨ ا ع ا ه ي ن ا ط ي ر ب ل ا ه ي ن د ر ا ل ا ه د ه ا ع ا م ل ل س ي ئ ر ل ا ف د ه ل ا ن ا ك — Jordan civics HS: primary goal of the 1948 Jordanian-British treaty — Jordanian national framing
D23	Severely garbled OCR text making question almost illegible

D24	Egypt university political science with severe OCR corruption — content partially unreadable
D14	تارمع عرباً؟ ملسو هيلع هلل يلص يبنل رمتعا ةرم مك / مالسل هيلع نامي لس ؟ يطل قطنم ملع يذل يبنل وه نم — Pan-Islamic knowledge questions: no country field — factual hadith/Quran questions
D35	Arabic language reading with Islamic ritual context — assumes Muslim reader perspective
D4	ل دُع ...يندرأال يبرعل شي جل تال ي كشت نم ةيبرعل ا ةوقل... 1952 ماع ندرأال ي ةيفال تئ ةموكح لوأ لي كشت مت — All 5 sampled Civics Middle School items are exclusively about Jordanian political history

Strengths

- Ground-up data collection from authentic school exams in eight countries, distinct from prior translation-based approaches [Q1, Q5, Q6, Q26]
- Palestine confirmed as top-3 source country with substantive curriculum content across multiple subjects [WEB-3, WEB-4, DATASET-D17, DATASET-D20, DATASET-D21]
- Native-speaker workforce with regional collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia [Q31, Q96]
- GPT-4 performance gap (80% on translated MMLU vs 72.5% on ArabicMMLU) supports construct validity for culturally grounded knowledge [Q64, Q65]

ID	Evidence
Q1	“we present ArabicMMLU, the first multi-task language understanding benchmark for the Arabic language, sourced from school exams across diverse educational levels in different countries spanning North Africa, the Levant, and the Gulf regions.” (p.1)
Q5	“Although large language models (LLMs) such as GPT-3.5 (Ouyang et al., 2022), BLOOMZ (Muennighoff et al., 2022), and Jais (Sengupta et al., 2023) have been pretrained with substantial coverage of Modern Standard Arabic (MSA), their reasoning and knowledge assessments are primarily conducted using datasets translated from English to Arabic (Sengupta et al., 2023; Huang et al., 2023), which means there is limited capacity to evaluate content specific to Arabic.” (p.1)
Q6	“This reliance on translation systems not only demonstrates an Anglocentric approach (Ramesh et al., 2023; Talat et al., 2022) but also potentially introduces errors and biases.” (p.1)
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WEB-4	HuggingFace dataset card confirms Palestine major contributor (https://huggingface.co/datasets/MBZUAI/ArabicMMLU)
D17	ركذ ام عي مج / نم لس / اينامست / ادنل يزوي ن :ايسوناي أقأ ةراقب ةطيحم ل رزجل نم — Palestine middle school geography: world geography (Oceania islands) — no regional specificity
D20	يئ انثلا ماظنل :اهتجل اعمو تاناي بل اظفحو لي ثمت ي ف بوسا حل هم دختسي دعل ةمظنأ دحأ — Palestine primary CS: binary number systems — STEM, no contested content
D21	تادي بيلل عي نصت نع ةل وئسم ةيل اتل تايضعل ا يدح / X100 :وه يئ وضرل ره مجل ا ي ةي تي زل ا ةسدعل ري بكت رادقم — Palestine HS biology: standard cellular biology — culturally neutral STEM
Q31	“The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian).” (p.4)
Q96	“We extend our gratitude to all collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia who participated in the data collection process.” (p.10)
Q64	“It is noteworthy that in the GPT-4 technical report (OpenAI, 2023), the accuracy of the English-Arabic translated MMLU dataset is reported as 80%, which is 8 points higher than our ArabicMMU results.” (p.7)
Q65	“One possible explanation for this difference is that our ArabicMMU presents a greater challenge due to its inclusion of a higher proportion of Arabic-specific content.” (p.7)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?

IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — many inputs require region-specific knowledge of Arab history, geography, Jordanian/Egyptian/Palestinian curricula, and Islamic religious practice [Q72, Q73, DATASET-D1, DATASET-D14].

IC-2: Partially — pan-Arab framing is acceptable per elicitation [A2], but contested historical content (e.g., King Abdullah I and Zionism [DATASET-D1], 1948 events) is framed from Jordanian curriculum perspective without flagging this country-specificity, conflicting with the deployment requirement to flag rather than silently generalize country-specific content.

IC-3: Limited Western-specific content concern — some Geography items ask about non-Arab global facts (Kilimanjaro, Shanghai) [MINOR Concern 11], but these are general world knowledge rather than Western-specific cultural content. The more pressing issue is Jordan-specific content presented as default Arab content.

IC-4: Verified by 2 native Arabic speakers on 100 sampled questions [Q42], yielding 96% match [Q43] — but no Moroccan, Palestinian, or Kuwaiti annotators participated in collection or verification [Q31, Q96]. Palestinian-sourced contested content was verified by non-Palestinian workers.

IC-5: Documented content issues: (1) Jordan-skew (~6,000+ questions) vs. ~100 for others [Q89]; (2) Morocco's school-curriculum content largely absent [DATASET-D5]; (3) Kuwait effectively absent [Limitations.5]; (4) OCR corruption in university Egyptian items [DATASET-D23, DATASET-D24]; (5) Islamic Studies presented prescriptively for Muslim students rather than explanatorily for non-Muslim visitors [DATASET-D14, DATASET-D35]; (6) Jordanian-curriculum framing of contested historical events [DATASET-D1, DATASET-D2, DATASET-D3].

ID	Evidence
Q72	"We present the performance of open-source models on selected subjects that potentially contain Arabic-specific contexts." (p.7)
Q73	"These subjects include history, geography, civics, political" (p.7)
D1	دحو مامت: ني طس ل ف ي ه اع ام طأ و ة ني نو ي ه ص ل ا ة ك ر ح ل ا ن م ل و ا ل ا ه ل ا د ب ع ك ل م ل ا ف ق ا و م ن م ت س ي ل ة ي ت ا ل ا ن م ة د ح و ا م ص ف ر / ن ي ن ي ط س ل ف ل ا ة م ا ل ا ق و د ن ص م ع د ي ل ا ب ر ع ل ا ة و ع د / ق ا ر ب ل ا ط ئ ا ح ي ل ا ا ر م د و ه ي ل ا ء ا ط ع ا م ص ف ر / م 1950 م ا ع ن ي ت ف ض ل ا ا ع ا ط ا ق ا ص ف ر ر و ف ل ب د ع و و ك ي ب س ك ي ا س ة د ه ا ع — Jordan HS history question about King Abdullah I's positions on Zionism and Palestine — framed from Jordanian curriculum perspective
D14	— ت ا ر م ع ب ر ا ؟ م ل س و ه ي ل ع ل ل ا ي ل ص ي ب ن ل ا ر م ت ع ا ة ر م ك / م ل س ل ا ه ي ل ع ل ا م ي ل س ؟ ر ي ط ل ا ق ط ن م م ل ع ي ذ ل ا ي ب ن ل ا و ه ن م — Pan-Islamic knowledge questions: no country field — factual hadith/Quran questions
Q42	"Finally, we assessed the accuracy of our data collection by having two native Arabic speakers annotate 100 randomly sampled questions. They were provided with all metadata, including the answer key, and tasked with verifying the correctness of each sample using any available resources (e.g., search engines)." (p.4)
Q43	"We found that 96% of the questions and answer keys match on average, while the remaining could have incorrect answer keys." (p.4)
Q31	"The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4)
Q96	"We extend our gratitude to all collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia who participated in the data collection process." (p.10)
Q89	"ArabicMMLU does not represent all Arabic countries equally. For example, we have collected over 6K multiple-choice questions from Jordan, while other countries are represented with only 100 questions or, in some cases, not at all." (p.9)
D5	ي ن ط و ل ا ع ا ف د ل ا ب ة ف ل ك م ل ا ة ي م و ك ح ل ا ط ل س ل ا و ل د ع ل ا ر ي ز و ن م ... ف ا ن ت س ا ل ا ل ج ا ة ت ب ي — All 5 sampled Law (Professional) items are from Morocco (Moroccan procedural law)
D23	د ر د ... ص ا خ ت ل ا ل ا ن ي ت ب ة ط ل ت د ل ا ع ت ي ي ه ت ة ت ج ر د — Severely garbled OCR text making question almost illegible
D24	ق ب س ا م ل ك ؟ م ا ر ر س ا ل ا ر ر ل و د ل ا ر ر ف م ك ا ر ر ل ا ر ر ف ا ر ر ر ف ا و ت ب ر ر ر ر ر ت ل ا ر ر س ا ل ا ط و ر ر ر ر ر ش ل ا ن ر ر م — Egypt university political science with severe OCR corruption — content partially unreadable

D35	أعرض نودب: [غارف] ملسم اليلصي نأ نكمي ال — Arabic language reading with Islamic ritual context — assumes Muslim reader perspective
D2	نب لصيل في أم أ-٢ / نيسحل النب هلل ادبع ري أم أ-١: سيرا ب في ءافلحل هديق ذلحلصل رمؤم في برعلل ____ لثم لال ط نب نيسحلل-٤ / لي نب نيسحلل في رشلل-٣ / نيسحلل — Jordan civics HS: who represented Arabs at Paris Peace Conference — Jordanian Hashemite framing
D3	بادتالال كف دويق نم دالبل ررحت: ١٩٢٨م عهناطيربلل هيندرأال هءاعملل سيئرلل فءهلل ناك — Jordan civics HS: primary goal of the 1948 Jordanian-British treaty — Jordanian national framing

Information Gaps

- Exact per-country question counts beyond top-3 not published
- Proportion of OCR-corrupted items in full dataset

Requires Expert Verification

- Whether ground-truth labels for Palestinian-sourced contested-history questions reflect Palestinian historiography
- Whether Moroccan curriculum perspectives in history/geography are present in the full dataset beyond the sampled configs
- Whether Islamic Studies items would be considered appropriately framed for non-Muslim educational use

Remediation

Gap 1: Morocco's school-curriculum content (history/geography/civics) is largely absent; Kuwait is effectively absent [DATASET-D5, Limitations.5]

Recommendation: Commission targeted collection of Moroccan and Kuwaiti primary/middle/high school exam items in history, geography, and civics, with native annotators from each country, to extend country coverage before deployment-relevant evaluation.

ID	Evidence
D5	ين طول عافدل اب ءفل كم ال ءيموكحل ءطلسل او لدعل ريزو نم... فانئستالال ل ءءتبـي (Professional) items are from Morocco (Moroccan procedural law)